

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

B.Sc. (IT) Semester-I

University Examination March-2018

CA 402 Introduction to Programming in C

Date: 26.03.2018, Monday Time: 10.00 am to 01.00 pm Maximum Marks: 70

Instructions:

1. Attempt both sections in separate answer books.
2. The figure to the right indicates marks.
3. Make a suitable assumption when necessary.

SECTION-I

Q.1 Attempt the following. [11]

1. Define: Algorithm
2. To take the values from the user, _____ statement is used in a program.
(a) printf() (b) scanf() (c) clrscr() (d) None of the above
3. C was developed by _____.
(a) Dennis Ritchie (b) Kenn Thompson (c) Bjarne Stroustrup
4. _____ operator is used to find remainder of an integer division.
(a) / (b) * (c) % (d) &
5. printf () function is available in _____ header file.
(a) stdio.h (b) string.h (c) math.h (d) conio.h
6. There must be one end in a flowchart. True / False
7. _____ symbol is used to represent condition in a flowchart.
(a) rectangle (b) circle (c) arrow (d) diamond
8. Loop is used to repeat the execution of list of statements. True / False.
9. _____ are reserved words whose meaning can not be changed.
(a) variable (b) keywords (c) identifiers (d) none of the above
10. Comment is an executable statement. True / False
11. Execution of a program begins with _____.
(a) comment (b) header files (c) main() (d) none of the above

Q.2 [A] Explain all the characteristics of C programming language. [06]

[B] Explain with diagram structure of C program. [06]

OR

Q.2 [A] Write an algorithm to find summation of two numbers entered by user. [06]

[B] Explain all the symbols used in flowchart. [06]

Q.3 Answer following questions in detail (Any three) [12]

1. What is called Identifier? Explain all the rules of identifiers.
2. What is called constant? Explain types of constants.
3. Draw a flowchart to find the largest of three numbers.
4. Explain with diagram compilation and execution of C program.
5. What is called loop? Explain types of loop with example.

[P.T.O.]

SECTION-II

Q.4 Attempt the following.

[11]

- Comparison can be done with the help of _____ operator.
(a) relational (b) arithmetic (c) logical (d) conditional
- _____ symbol is used as an assignment operator.
(a) % (b) / (c) = (d) &
- What will be the value of a and i after following code execution?
i = 5;
a = ++i;
(a) i=5, a=5 (b) i=5, a=6 (c) i=6, a=5 (d) i=6, a=6
- In _____ loop, the body of the loop is always executed at least once.
(a) while (b) for (c) do...while (d) none of the above
- Array is a derived data type. True / False
- If initializing elements of an integer array are less than the size of an array, the remaining elements will set to _____.
(a) NULL (b) 0 (c) 2 (d) 1
- _____ is a sequence of characters enclosed in a pair of quotes.
(a) string (b) variable (c) comment (d) loop
- _____ function in C language concatenates two given strings.
(a) strlen() (b) strcmp() (c) strcpy() (d) strcat()
- _____ specifies the type of value that the function is expected to return to the calling function.
(a) function name (b) function arguments
(c) function return type (d) function body
- By default, function will return void value. True / False
- The parameter names must be the same in the prototype declaration and the function definition. True / False

Q.5 [A] Explain in detail all the operators available in C.

[06]

[B] Explain in detail categories of functions.

[06]

OR

Q.5 [A] Which are the elements of user defined function? Explain in detail.

[06]

[B] Write a program to find average of n numbers using an array.

[06]

Q.6 Answer following questions in detail (Any three)

[12]

- What is called type conversion? Explain in detail with its types and example.
- Write down rules of switch statement.
- Write a program to read the marks of 10 students using array and display total on screen.
- What is called array? Explain its type.
- Write a C program to swap two numbers using function.